FOUNDATION OF HUMAN COMPUTER INTERACTION

WHAT IS HCI

Human–Computer Interaction (HCI) is the study of how people interact with technology. It examines both human capabilities and computer capabilities to ensure that systems are designed to be effective, efficient, and satisfying to use. In simple terms, HCI asks: “How do we design technology that works well for humans?”

Understanding HCI is essential because poorly designed systems can lead to errors, frustration, and inefficiency—while well-designed systems improve productivity, safety, and user experience.

HUMAN CAPABILITIES

To design successful interfaces, we must begin by understanding the user. Humans have strengths and limitations, and both influence how they interact with computers.

- **Cognitive Capabilities**
- **Physical Capabilities**
- **Sensory Capabilities**

COGNITIVE CAPABILITIES

Perception:

Perception refers to how humans interpret sensory information—what we see, hear, and feel.

Poor perception leads to poor interaction. For example:

- Text that is too small is hard to read.
- Low contrast makes buttons hard to see.
- Distracting colors make information difficult to process.

COGNITIVE CAPABILITIES

Attention

Attention is limited. Humans cannot focus on too many elements at once.

Design implication:

- Reduce clutter.
- Highlight important elements.
- Provide clear visual hierarchy.

COGNITIVE CAPABILITIES

Memory

Memory is divided into three types:

Sensory memory

– Lasts only a few seconds.

Short-term/Working memory

– Very limited capacity (7 ± 2 items on average).

– Interfaces should avoid forcing users to remember too much.

Long-term memory

– Holds large amounts of information, but slow to recall.

– Good interfaces rely on recognition (icons, menus) rather than recall (commands).

PHYSICAL CAPABILITIES

Motor Movement

People cannot perform very precise or fast movements consistently. Good design includes:

- Buttons large enough to tap
- Minimal need for very accurate pointing
- Swipe gestures instead of typing long text

PHYSICAL CAPABILITIES

Fitts' Law

The time to reach a target depends on its size and distance.

This is why critical buttons (e.g., Submit, Save) are made larger.

PHYSICAL CAPABILITIES

Reaction Time

Humans need time to react to changes.

Systems must avoid sudden or confusing transitions without feedback.

SENSORY CAPABILITIES

Vision

The primary sense used in computer interaction.

Design rules:

- Use readable fonts
- Avoid relying only on color to convey meaning
- Maintain good contrast
- Use clear, simple visual layouts

SENSORY CAPABILITIES

Hearing

Used in alerts, notifications, and feedback.

Not everyone can hear well, so audio should always have a visual alternative.

SENSORY CAPABILITIES

Touch

Touchscreens rely on haptic feedback and finger gestures.

Design considerations:

- Finger-friendly spacing
- Gesture consistency
- Avoid hidden controls

COMPUTER CAPABILITIES

Just as humans have capabilities, computers also have their strengths and limitations.

Input Device
Output Device
Computer Strength
Computer Limitation

INPUT DEVICE

These allow the user to send information to the computer.

Examples:

- Keyboard
- Mouse
- Touchscreen
- Microphone (voice input)
- Sensors (GPS, accelerometer)
- Cameras (facial recognition)

Input methods must be matched to the task and user needs.

OUTPUT DEVICE

These deliver information from the system to the user.

Examples:

- Monitor / Display
- Speakers
- Printers
- Haptic devices (vibration in phones or game controllers)

Good output design ensures clarity and immediate feedback.

COMPUTER STRENGTHS

Computers excel at:

- Speed – Fast calculations
- Accuracy – Precise operations
- Storage – Large memory capacity
- Consistency – Perform tasks predictably

COMPUTER LIMITATIONS

Computers cannot:

- Understand context like humans
- Interpret emotions accurately
- Deal with ambiguity without data
- Make ethical judgments

This is why interface design must bridge human limitations and computer inflexibility.

THE INTERACTION

HCI centers on the interaction between the user and the system.

The Interaction Loop

Interaction Styles

THE INTERACTION LOOP

This loop explains how users interact with technology:

1. Form goal
2. Plan action
3. Execute action
4. System processes input
5. System outputs results
6. User interprets feedback

If any step breaks, the user becomes frustrated.

INTERACTION STYLES

Different systems use different interaction styles:

Command Line Interface (CLI)

- **Users type commands**
- **Very powerful but requires memory**
- **Used by programmers and system administrators**

INTERACTION STYLES

Menu-Based Interaction

- Lists of options
- Easy to use, ideal for beginners

Graphical User Interface (GUI)

- Windows, icons, menus, pointers
- Visual and intuitive

Direct Manipulation

- Dragging, resizing, rotating objects
- Example: Touchscreen gestures

INTERACTION STYLES

Form-Based Interaction

- Users input data in structured fields
- Used in applications like registration forms

Conversational Interfaces

- Chatbots, voice assistants (Siri, Alexa)
- Natural language interaction

Virtual/Augmented Reality

- Immersive environments
- Used in gaming, training, simulation

HCI PARADIGMS

HCI paradigms are the major shifts in how humans interact with computers.

Batch Processing Era

- No direct interaction
- Users submitted jobs and waited for results
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Command-Line Era

- Text-based interaction
- Users typed commands
- Steep learning curve

HCI PARADIGMS

Graphical User Interface (GUI) Era

- Introduced by Xerox PARC, popularized by Apple and Microsoft
- WIMP model: Windows, Icons, Menus, Pointer
- Easier for average users

Mobile & Wearable Paradigms

Examples:

- Smartphones
- Smartwatches
- Fitness trackers

Focus: Anytime, anywhere access.

HCI PARADIGMS

Immersive Paradigms

Virtual Reality (VR)

- Fully simulated environment.

Augmented Reality (AR)

- Digital elements layered over the real world.
- (Example: Pokémon GO)

Mixed Reality (MR)

- Blend of real and virtual environments.

Understanding human capabilities, computer capabilities, the interaction process, and evolving HCI paradigms is essential in designing modern, user-centered systems. Good HCI improves usability, minimizes errors, and enhances user satisfaction.

A well-designed system is not only functional—it is intuitive, responsive, and pleasant to use.

THANK YOU